Scaling Verification Depth for Stronger Reasoning

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October 2026

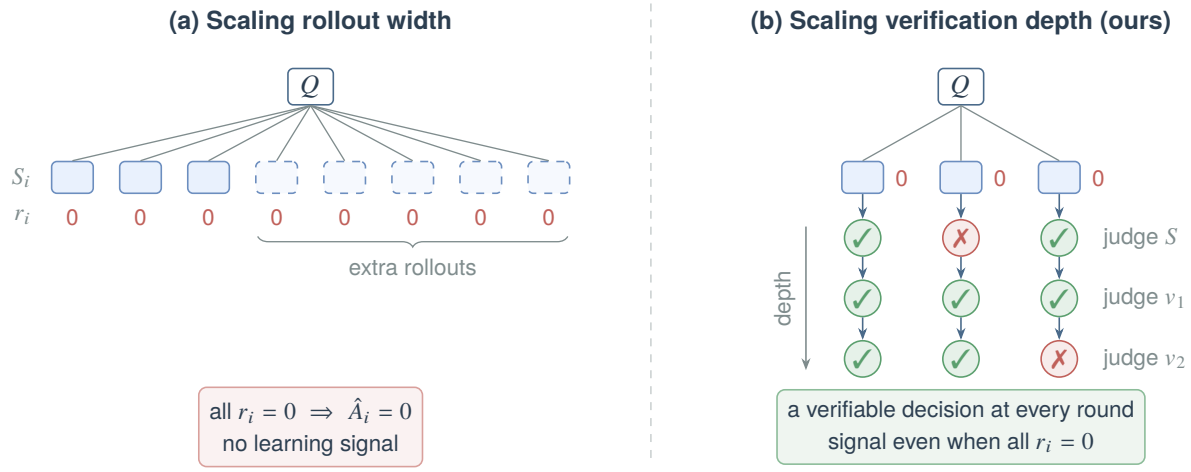


Figure 1: Two ways to spend extra training compute on a hard question. (a) Scaling rollout width samples more solutions, but if all of them fail, every reward is 0 and the group-relative advantages vanish. (b) Scaling verification depth (ours) verifies each existing solution over several rounds, each judging the previous verdict.

1 Motivation

Recent advances in reinforcement learning with verifiable rewards (RLVR) have increasingly focused on improving training efficiency by scaling the number of generated solutions. Methods such as VIP [1] and DARS [2] adaptively allocate rollout budgets across questions to improve exploration and learning from difficult problems. However, scaling rollout width has an inherent limitation: generating more solutions does not necessarily provide more useful learning signals. For challenging questions, additional rollouts may still fail to produce correct answers, resulting in uniform zero rewards and vanishing group-relative advantages. Moreover, existing approaches primarily invest additional computation in exploring new solutions, while overlooking the opportunity to extract richer learning signals from already generated attempts. This raises an important question: rather than continuously generating more solutions, can we improve reasoning by learning more deeply from each existing solution?

Motivated by this observation, we propose an alternative scaling direction: **verification depth** (Fig. 1). Instead of allocating additional computation to solution generation, *we introduce a sequence of verification rounds in which the model progressively reassesses its previous judgments.* Each round produces a verifiable decision, allowing the model to receive additional learning signals even when the original solution is incorrect. Importantly, our recursive verification mechanism encourages the model not only to identify incorrect solutions but also to recognize mistakes in its own judgments and preserve correct assessments. By optimizing these decisions across multiple verification depths, our method aims to extract more supervision from each generated solution while strengthening the model's ability to evaluate and reconsider its reasoning. We hypothesize that this approach can improve reasoning performance more efficiently than simply increasing rollout width under a comparable training budget.

2 Method

2.1 Preliminaries

Policy. Let π_θ be an autoregressive language model with parameters θ. Given a question Q, it generates an output $o = (o_1, \ldots, o_{|o|})$ with probability

$$\pi_\theta(o \mid Q) = \prod_{t=1}^{|o|} \pi_\theta(o_t \mid Q, o_{<t}). \tag{1}$$

Reinforcement learning with verifiable rewards. Let $\mathcal{D}$ be a dataset of pairs (Q, R), where R is the reference answer. A rule-based verifier compares the final answer in an output o with R and assigns a binary reward

$$r(o, R) = \text{Verify}(o, R) \in \{0, 1\}.$$

RLVR maximizes $\mathbb{E}_{(Q,R)\sim\mathcal{D},\, o\sim\pi_\theta(\cdot|Q)}\big[r(o, R)\big]$.

Group relative policy optimization. For each question Q, GRPO samples a group of G outputs $\{o_i\}_{i=1}^G$ from the old policy $\pi_{\theta_{\text{old}}}$ and computes their rewards $\{r_i\}_{i=1}^G$. Let $o_{i,t}$ denote the t-th token of o_i, for $t = 1, \ldots, |o_i|$. All tokens of o_i share the same group-normalized advantage

$$\hat{A}_{i,t} = \frac{r_i - \text{mean}\left(\{r_j\}_{j=1}^G\right)}{\text{std}\left(\{r_j\}_{j=1}^G\right)}. \tag{2}$$

The policy is updated by maximizing

$$\begin{aligned}\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{(Q,R)\sim\mathcal{D},\, \{o_i\}_{i=1}^G\sim\pi_{\theta_{\text{old}}}(\cdot|Q)}\Bigg[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(\min\big(\rho_{i,t}(\theta)\hat{A}_{i,t},\\ \text{clip}\left(\rho_{i,t}(\theta), 1-\varepsilon, 1+\varepsilon\right)\hat{A}_{i,t}\big) - \beta\,\mathbb{D}_{\text{KL}}\big[\pi_\theta \,\|\, \pi_{\text{ref}}\big]\Big)\Bigg],\end{aligned} \tag{3}$$

where

$$\rho_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid Q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid Q, o_{i,<t})} \tag{4}$$

is the importance ratio, ε is the clipping range, and β weights the KL penalty toward a frozen reference policy π_{ref}.

2.2 Proposed Idea: Scaling Verification Depth

Each training step has two rollout phases (Fig. 2). Phase 1 is the standard GRPO rollout: the policy samples G solutions to a question. Phase 2 scales verification in depth: the policy generates one verification chain of M sequential rounds for each solution, where M denotes the verification depth. Both phases sample from $\pi_{\theta_{\text{old}}}$ and are optimized in the same update. Each verification round is trained as a separate prompt-response example with its own correctness reward.

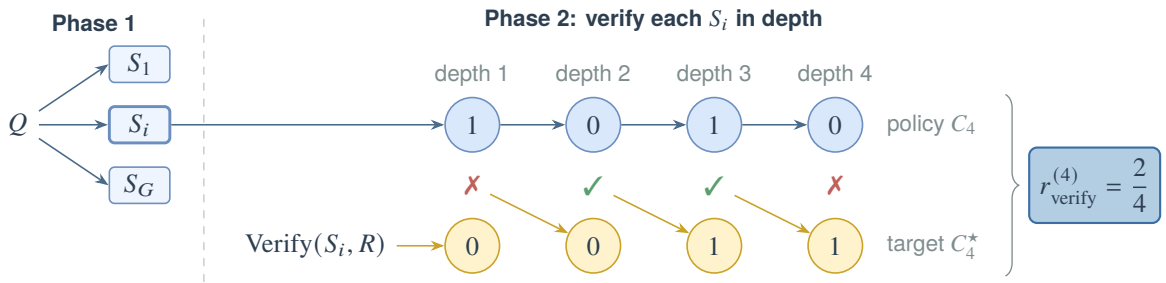


Figure 2: Two rollout phases. Phase 1 samples G solutions to Q. Phase 2 generates one verification chain of depth M for each S_i, illustrated with $M = 4$. Here S_i is incorrect. Blue: policy verdicts; v_1 judges S_i, and each v_d judges v_{d-1}. Yellow: target verdicts. Each match (✓/✗) determines the next target and supplies the current round's reward. The box reports the average accuracy across depths.

2.2.1 Phase 1: Solution Generation.

For each question Q, the policy samples G solutions $\{S_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid Q)$, and each receives the outcome reward $r_i = r(S_i, R)$. These rollouts are trained with $\mathcal{J}_{\text{GRPO}}$ in (3), with $o_i = S_i$.

2.2.2 Phase 2: Verification Chain Generation.

Verification chain. For each solution S_i, the policy generates one verification chain of depth M. We describe one chain for a fixed pair (Q, S) and drop the indices.

At depth 1, the policy receives the question and the solution. At depth $d \geq 2$, it additionally receives only the previous round's final answer A_{d-1}. The context is

$$x_d = \begin{cases} (Q, S), & d = 1, \\ (Q, S, A_{d-1}), & d = 2, \ldots, M, \end{cases} \tag{5}$$

and the policy generates a response

$$V_d \sim \pi_{\theta_{\text{old}}}(\cdot \mid x_d), \qquad \bar{A}_d = \text{ExtractAnswer}(V_d), \tag{6}$$

where $\bar{A}_d$ is the final answer after the closing </think> tag and contains a binary verdict v_d. The reasoning trace inside the <think> tags is not passed to subsequent rounds.

At depth 1, the verdict judges the solution: $v_1 = 1$ claims that S is correct. At depth $d \geq 2$, the verdict judges the previous verdict: $v_d = 1$ claims that v_{d-1} is correct. To make the final answer self-contained, we attach a running conclusion $c_d \in \{0, 1\}$ about whether S is correct:

$$c_1 = v_1, \qquad c_d = \begin{cases} c_{d-1}, & v_d = 1, \\ 1 - c_{d-1}, & v_d = 0, \end{cases} \quad d = 2, \ldots, M. \tag{7}$$

The forwarded final answer is $A_d = (\bar{A}_d, c_d)$. The field c_d is computed deterministically from the sampled verdict and the previous conclusion, without using the reference answer. It summarizes the verdict history without retaining any earlier reasoning traces. The policy chain is

$$C_M = (v_1, \ldots, v_M). \tag{8}$$

Target chain. The target chain $C_M^\star = (v_1^\star, \ldots, v_M^\star)$ holds the correct verdict at every depth. The first target is the phase-1 outcome reward of S. For $d \geq 2$, the verdict v_{d-1} is correct exactly when it equals its own target,

so

$$v_1^\star = \text{Verify}(S, R), \qquad v_d^\star = \mathbb{1}\big[v_{d-1} = v_{d-1}^\star\big], \quad d = 2, \ldots, M. \tag{9}$$

Let $m_d = \mathbb{1}[v_d = v_d^\star]$ indicate a match at depth d. The recursion then reads $v_{d+1}^\star = m_d$: the target at each depth records whether the policy was right one depth earlier. The target chain is computed after the policy chain is generated and is never shown to the policy.

Round reward. Each verification round receives its own binary correctness reward:

$$r_d^{\text{verify}} = \mathbb{1}\big[v_d = v_d^\star\big] = m_d \in \{0, 1\}. \tag{10}$$

Equivalently, a verdict is correct exactly when its running conclusion agrees with the solution's outcome:

$$r_d^{\text{verify}} = \mathbb{1}\big[c_d = \text{Verify}(S, R)\big]. \tag{11}$$

The average across depths is retained as a summary score:

$$r_{\text{verify}}^{(M)} = \frac{1}{M} \sum_{d=1}^{M} r_d^{\text{verify}} \in [0, 1]. \tag{12}$$

Training uses the individual round rewards rather than assigning this average to every round.

2.2.3 Training Objective.

Round-level feedback. For solution S_i, let $x_{i,d}$, $V_{i,d}$, and $r_{i,d}^{\text{verify}}$ denote the context, response, and reward at depth d. Each pair $(x_{i,d}, V_{i,d})$ is a separate training example. Its sampled context is held fixed during the update, and its feedback weight is

$$\hat{A}_{i,d}^{\text{verify}} = r_{i,d}^{\text{verify}} - \frac{1}{2} = \begin{cases} +\frac{1}{2}, & v_{i,d} = v_{i,d}^\star, \\ -\frac{1}{2}, & v_{i,d} \neq v_{i,d}^\star. \end{cases} \tag{13}$$

The fixed baseline $1/2$ gives positive feedback to correct rounds and negative feedback to incorrect rounds, without comparing different branches or depths. Phase 2 therefore uses clipped policy gradient with a fixed baseline rather than group-normalized GRPO.

Verification objective. Let $V_{i,d,t}$ denote the t-th generated token of $V_{i,d}$. The importance ratio is

$$\rho_{i,d,t}^{\text{verify}}(\theta) = \frac{\pi_\theta(V_{i,d,t} \mid x_{i,d}, V_{i,d,<t})}{\pi_{\theta_{\text{old}}}(V_{i,d,t} \mid x_{i,d}, V_{i,d,<t})}. \tag{14}$$

The policy maximizes the round-level clipped objective

$$\begin{aligned} \mathcal{J}_{\text{verify}}(\theta) = \mathbb{E}\Bigg[\frac{1}{GM} \sum_{i=1}^{G} \sum_{d=1}^{M} \frac{1}{|V_{i,d}|} \sum_{t=1}^{|V_{i,d}|} \Big(\min\big(&\rho_{i,d,t}^{\text{verify}}(\theta) \hat{A}_{i,d}^{\text{verify}}, \\ &\text{clip}\big(\rho_{i,d,t}^{\text{verify}}(\theta), 1-\varepsilon, 1+\varepsilon\big) \hat{A}_{i,d}^{\text{verify}}\big) - \beta\, \mathbb{D}_{\text{KL}}\big[\pi_\theta \,\|\, \pi_{\text{ref}}\big] \Big) \Bigg], \end{aligned} \tag{15}$$

where the expectation is over dataset questions, sampled solutions, and verification chains. The KL term is evaluated at each token's conditional context. Each round's feedback is shared by all tokens of its generated response, including its reasoning and final answer. The question, solution, previous final answer, and round

instructions are masked from the loss. The deterministic conclusion field is not generated by the policy and receives no policy loss.

This objective trains each round for its own verdict accuracy on the sampled contexts; rewards from later rounds are not assigned to earlier responses. The complete verification chain is not concatenated into a single training response. The policy maximizes

$$\boxed{\mathcal{J}(\theta) = \mathcal{J}_{\text{GRPO}}(\theta) + \mathcal{J}_{\text{verify}}(\theta)}. \tag{16}$$

The verification rewards evaluate binary verdicts rather than the validity of their reasoning traces. Phase 2 introduces $G \times M$ verification rounds and the same number of training examples per question.

References

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[2] Zhicheng Yang, Zhijiang Guo, Yinya Huang, Yongxin Wang, Dongchun Xie, Hanhui Li, Yiwei Wang, Xiaodan Liang, and Jing Tang. Depth-breadth synergy in rlvr: Unlocking llm reasoning gains with adaptive exploration. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 144643–144669. PMLR, 2026. URL https://proceedings.mlr.press/v306/yang26s.html.